VIT

Vellore Institute of Technology

Reg. Number:	22BAI1265

Continuous Assessment Test (CAT) – I AUGUST 2025

Programme	:	B.Tech. Computer Science and Engineering (AI/ML)	Semester	:	Fall Semester 2025 – 26
Course Code & Course Title	:	BCSE416L & Game Programming	Class Number	:	CH2025260102071
Faculty	:	S. Graceline Jasmine	Slot	:	C1+TC1
Duration	:	1hr & 30mins	Max. Mark		50

General Instructions:

- Write only your registration number on the question paper in the box provided and do not write other information
- Use statistical tables supplied from the exam cell as necessary
- Use graph sheets supplied from the exam cell as necessary
- Only non-programmable calculator without storage is permitted

Answer all questions

Q. No	Sub Sec.	Description	Marks	CO	BT Level
1.		You have been appointed as the Lead Game Designer for a futuristic 3D sci-fi survival game set in a Martian colony. The environment features vast alien landscapes, dust storms, underground tunnels, and advanced alien technology. The aim is to create a visually immersive experience that supports environmental storytelling.		1	K1
	a)	Based on the game's setting and the need for high visual immersion, identify three key art and visual roles you would include in your design team. Justify how each role is essential for creating believable environments and enhancing the player's experience.	5		
	b)	Identify two major visual or aesthetic challenges that might arise when developing a realistic Martian environment. Suggest how the team you proposed can address these challenges effectively.	5		
2.	a)	In an RPG, players can pick up weapons, potions, and quest items of varied sizes. The inventory must handle frequent additions, removals, and lookups efficiently while conserving memory, especially for mobile or VR platforms with limited resources. Identify the most memory-efficient data structure you would use to manage the inventory and justify your choice.	5	1	K3

	b)	Explain how your chosen data structure balances performance and memory usage for mobile or VR platforms.	5		
3.		A game object (a drone) in a 3D scene is currently at position P = (2, -1, 3) in world coordinates. The drone must rotate about the world in Y axis by 120° to face a new heading.		2	K3
	a)	Write the unit quaternion R_q that represents a rotation of 120° about the Y axis. Provide the steps and give the quaternion in the form q = (w, x, y, z).	5		
	b)	Using quaternion rotation, compute the new position P′ of the drone after applying the rotation to the point P = (2, -1, 3). Write the quaternion-vector rotation steps and provide the final coordinate position.	5		
4.		You are designing Puppy's Path to Recovery, a 2D side-scrolling Unity game where a puppy's actions reflect the player's rehabilitation progress. When the player performs exercises (mapped to keyboard input for testing), the puppy walks forward; when idle, it waits patiently. Completing exercises helps the puppy progress through a vibrant, supportive world, unlocking small rewards that encourage daily participation.		2	K4
	a)	List the steps required in Unity to set up the Idle and Walk animations for the puppy sprite, including Animator configuration and transitions.	5		
	b)	Write a C# script to control movement and animation so that pressing left/right keys moves the puppy accordingly and plays the correct animation state	5		
5.	a)	You are building a real-time lighting system in Unity for a sunset forest scene. The system must calculate what the camera sees to generate accurate shadows and reflections. Which ray casting approach would you choose to calculate visibility from the camera's perspective, and why is it best suited for this task?	5.	1	K2
	b)	In a medieval castle scene, a glowing crystal lantern above the main gate radiates light evenly in all directions, while a rotating beacon sends a concentrated cone of light across the courtyard. Identify the most suitable type of light for the lantern and for the rotating beacon, and explain why each is the best match for the desired light behaviour.	5		

**********All the best **********

R Radhika
13/8/25